

Intelligent Vision-Based Manual / Automatic Robotics Manipulator (*A landmine Detection Robot*)



Author

Imran Nazir
F22BINCE1M02021

Muhammad Zulqarnain
F22BINCE1M02001

Waqas Ajaz
F22BINCE1M02006

Supervisor
Dr. Zain Al Abiden Akhtar
Assistant Professor

DEPARTMENT OF INFORMATION AND COMMUNICATION
ENGINEERING
FACULTY OF ENGINEERING AND TECHNOLOGY,
THE ISLAMIA UNIVERSITY OF BAHAWALPUR,
PAKISTAN
June 2026

Intelligent Vision-Based Manual / Automatic Robotics
Manipulator
(A Landmine Detection Robot)

Author

Imran Nazir

F22BINCE1M02021

Muhammad Zulqarnain

F22BINCE1M02001

Waqas Ajaz

F22BINCE1M02006

A thesis submitted in partial fulfillment of the requirements for the degree of

BS Intelligent Systems and Robotics

Thesis Supervisor:

Dr. Zain Al Abiden Akhtar

Assistant Professor, Department of Information and Communication
Engineering

External Examiner Signature:

Thesis Supervisor Signature:

DEPARTMENT OF INFORMATION AND COMMUNICATION
ENGINEERING
FACULTY OF ENGINEERING AND TECHNOLOGY,
THE ISLAMIA UNIVERSITY OF BAHAWALPUR,
PAKISTAN

June 2026

ABSTRACT

Intelligent Vision-Based Manual / Automatic Robotics Manipulator

Imran Nazir

F22BINCE1M02021

Muhammad Zulqarnain

F22BINCE1M02001

Waqas Ajaz

F22BINCE1M02006

Thesis Supervisor:

Dr. Zain Al Abiden Akhtar
Assistant Professor

This thesis presents the design and development of an Intelligent Vision Based Manual / Automatic Robotic Manipulator for Landmine Detection and Manipulation Based on motion planning algorithm. The proposed system aims to reduce the risks associated with conventional landmine detection methods, which are often slow, dangerous, and highly dependent on human involvement. The system is developed as an ESP32-based mobile robotic platform integrated with a robotic manipulator arm, camera, GPS module, and metal detection sensor. The robot can manually operate by the user as well as autonomously navigate within a selected area using GPS guidance, detect metallic objects using the metal detection sensor, and safely retrieve them using its robotic arm without direct human intervention.

Keywords: Landmine Detection, Robotic Manipulator, ESP32, GPS-Guided Navigation, Metal Detection, Autonomous Robotics, Hazardous Environment Automation.

UNDERTAKING

I certify that research work titled “Intelligent Vision Based Manual / Automatic Robotics Manipulator” is my own work. The work has not been presented elsewhere for assessment. Where material has been used from other sources it has been properly acknowledged / referred.

Signature of Student

Imran Nazir

F22BINCE1M02021

Signature of Student

Muhammad Zulqarnain

F22BINCE1M02001

Signature of Student

Waqas Ajaz

F22BINCE1M02006

ACKNOWLEDGEMENTS

First of all, I would like to thank Allah Almighty, for giving me courage and strength to make this thesis successful.

*I am very thankful to my honorable supervisor, **Dr. Zain Al Abiden Akhtar**, for their guidance and support during the project.*

I am also very thankful to the faculty and staff of the Department of Information and Communication Engineering, The Islamia University of Bahawalpur, for providing the environment to make my research effectively. Lastly, I am grateful to my family and friends for their support and motivation

Table of Contents

ABSTRACT	ii
UNDERTAKING	iii
ACKNOWLEDGEMENT	iv
LIST OF FIGURES	xii
LIST OF TABLES	xiv
CHAPTER 1.....	1
Introduction	1
1.1 Background	1
1.2 Problem Statement	2
1.3 Objective of the Project	3
1.4 Scope of the project	4
1.4.1 Mechanical and Electrical Design	4
1.4.2 Sensing and Detection System	4
1.4.3 Control and Navigation	4
1.4.4 Human-Robot Interaction	4
1.4.5 Experimental Validation.....	4
1.5 Application	5
1.6 Significance of the study	5

1.7 Technological Framework	6
1.8 Social and Ethical Perspective.....	7
CHAPTER 2	9
Literature Review.....	9
2.1 Introduction.....	9
2.2 Challenges in landmine Detection	9
2.3 Conventional and Manual Demining Techniques	10
2.4 Robotics Systems for Mine Detection	10
2.5 Mobile Platform and UGVs	11
2.6 Sensor Integration and Detection Technology	11
2.7 GPS and IOT Integration	11
2.8 Sensor Fusion and Multimodal Detection.....	12
2.9 Machine learning and AI Approaches.....	12
2.10 Dataset and Algorithm Benchmarking.....	12
2.11 Manipulators and Robotics Arm in Demining	13
2.12 Navigation, Autonomy and Control System	13
2.13 Real-World Deployments and Limitations	13
2.14 Summery	14

CHAPTER 3	15
Hardware Design and Implementation	15
3.1 System Overview	15
3.2 System Architecture (Block Diagram)	15
3.3 Hardware Design (Circuit Diagram).....	16
3.4 Microcontroller and Processing Unit.....	17
3.4.1 ESP32 WROOM Module	17
3.4.2 ESP32-S3 Cam Module	18
3.5 Sensors and Actuators	20
3.5.1 Metal Detector	20
3.5.2 Camera	21
3.5.3 Buck Converter	21
3.5.4 DC Motor	22
3.5.5 Motor Driver	23
3.5.6 Servo Motors	23
3.5.7 GPS Module.....	24
3.5.8 PCA Board	24
3.5.9 QMC5883L Campas	25

3.5.10 Robotics Manipulator.....	25
3.5.11 Mobile Robot Platform	27
3.6 Power Supply Unit	27
3.6.1 Voltage Regulation Module	28
3.6.2 Power Requirement	29
3.6.3 Battery Management system (BMS).....	29
3.6.4 Charging and Backup.....	30
3.7 Software Integration and Control Algorithm	30
3.8 Mechanical Structure	30
3.9 Actuator and Motor Specification.....	31
3.9.1 Servo Motor Mapping (Angular Constraints).....	31
3.9.2 Drive System and Current Analysis.....	32
3.9.3 Current Requirement Summary	33
3.10 Material and Development Cost	33
3.11 Summery	34
CHAPTER 4	36
Software Implementation.....	36
4.1 Introduction.....	36

4.2 Desing Principles	36
4.3 FreeRTOS Framework	37
4.4 Software Functional Block	38
4.4.1 Navigation Module	38
4.4.2 Perception Module	38
4.4.3 Control Module.....	38
4.4.4 Manual Control	39
4.4.5 Autonomous Path Following	39
4.4.6 Web Interface	41
4.5 Integration and Communication	43
4.6 Implementation	43
4.7 Summary	44
CHAPTER 5	45
Testing and Performance Evaluation	45
5.1 Introduction	45
5.2 Purpose of Evaluation	45
5.3 Importance of Testing	46
5.4 Testing Procedure and Setup.....	46

5.4.1 Environment Configuration	48
5.4.2 Testing Phase and Categories.....	48
5.4.3 Test Case Scenarios.....	49
5.5 Performance Metrics	50
5.5.1 Data Acquisition Method	50
5.5.2 Performance Evaluation Criteria.....	51
5.5.3 Table of Test	52
5.6 Result	53
5.7 Graphical Representation.....	55
5.8 Advancement and Future Direction	56
5.8.1 Hardware Enhancement	56
5.8.2 Software Enhancement	56
5.8.3 User Interface and Interaction.....	57
5.8.4 Future Research Directions.....	57
5.9 Conclusion	57
CHAPTER 6	59
System Deployment and Feedback.....	59
6.1 Introduction.....	59

6.2 Deployment Process.....	59
6.2.1 Orientation and Awareness Campaign	60
6.2.2 Role of Assignment and Tsk Simulation.....	61
6.3 Human-Robot Interactions and Feedback.....	61
6.3.1 Feedback Table	62
6.4 Conclusion	63
CHAPTER 7	64
Conclusion and Future Directions	64
7.1 Conclusion	64
7.2 Result and System Evaluation	64
7.3 Challenges.....	65
7.4 System Scalability and Flexibility	66
7.5 Education Impact	66
7.6 Ethical and Social Consideration	66
7.7 Environmental Responsibility	67
7.8 Vision for Future Expansion	67
7.9 Final Remarks	68
Reference	69

List of Figure

Fig 3.1 Block Diagram.....	15
Fig 3.2 Circuit Diagram	16
Fig 3.3 ESP32 Devkit Module.....	18
Fig 3.4 ESP32-S3 Cam Module.....	19
Fig 3.5 Metal Detector	20
Fig 3.6 ESP32-Cam	21
Fig 3.7 Buck Converter.....	22
Fig 3.8 DC Motor for Wheel.....	22
Fig 3.9 L298 Motor Driver	23
Fig 3.10 Servo Motor.....	23
Fig 3.11 GPS Module.....	24
Fig 3.12 PCA Board.....	25
Fig 3.12 Campas	25
Fig 3.14 Robotics Manipulator	26
Fig 3.15 Robot Platform	27
Fig 3.16 Power Supply (Battery)	28
Fig 3.17 Mechanical Structure.....	31

Fig 4.1 Web Interface.....	41
Fig 4.2 web interface Manual Mode	42
Fig 4.3 Web Interface Automatic Mode e	42
Fig 5.1 Graph of Result.....	55

List of Table

Table 3.1 Power Requirement	29
Table 3.2 Servo Motor Angles	32
Table 3.3 Current Requirement	33
Table 3.4 Estimated Cost	33
Table 5.1 Performance Metrics	52
Table 6.1 Feedback	62

CHAPTER 1

Introduction

1.1 Background

Land mines and other unexploded metallic materials from past conflicts continue to be a major global concern across the world and risk of human and military person (R. Jawwharlal, Vol: 4(2), June 2025). Even decades after conflicts have encoded, these hidden devices remain buried in the ground, causing injuries, loss of life and long-term socio-economic problems. According to international humanitarian reports, 26,000 of civilians and soldiers are injured or killed every year due to accidental encounters by landmines with children representing 42% of these victims especially border area and in rural and post-war regions where agricultural and development activities are essential for survival (Singh Panwar et al., 2024)(Freese et al. (n.d.).

(R. Jawwharlal, Vol: 4(2), June 2025) Conventional mine detection and clearance techniques largely depend on manual inspection using handheld metal detectors and trained personnel. Although these methods are widely used, they are extremely time-consuming, physically exhausting and most importantly, highly dangerous. Human operators are directly exposed to the risk of accidental detonation, harsh environmental conditions and limited visibility. As a result, there is a growing global demand for automated and intelligent and reliable system that can perform detection and handling tasks while keeping humans at a safe distance.

(Heuschmid et al. (2025) Advancements in landmine robot system and sensor technologies have opened a new opportunity for developing autonomous machines and robotics development which capable of operating in hazardous environments. Microcontrollers such as ESP32 which

combined with GPS module, metal detector sensors and robotics manipulators enable the creation of a compact and powerful robotics platforms. These technologies make it possible to design systems that can navigate, sense, decide and act with minimal human intervention and this project utilizes these modern tools for a critical real-world safety challenge(Mukherjee et al., 2024).

1.2 Problem Statement

- Despite the ongoing detection efforts for landmine worldwide and existing methods are still suffered from several major limitations:
- There is a high risk to human life during detection and extraction processes and to reduce this risk, this project saves time and human life risk
- The slow and labor-intensive operations for detection especially in large or difficult areas where human access is difficult.
- Limited accuracy and resources for landmine detection due to which human face difficulty.
- Loss of life of civilian and soldiers during critical days in hard area like border and war zone.

The process of detecting mines is complex, dangerous and expensive. The main problem is not to remove mines, but to accurately determine the location of the mine. This entire operation is still performed manually, which leads to the fact that the pace of demining is very unsatisfactory(Sineglazov & Lesohorskyi, 2025a). Therefore, there is a strong need to develop a robot which is intelligent, low-cost and autonomous robotics system that can:

- Precisely scan a defined and user set area from the web interface.
- Detect metallic objects which may look like landmine.
- Safely manipulate the detected object and restart the operation.

- Provide accurate location information using GPS and send it to web server for monitoring and analysis of the environment with the help of Wi-Fi system.

This project aims to bridge the gap between safety, efficiency and affordability in landmine detection by introducing an automated robotics manipulator capable of performing these tasks with reduce human involvement.

1.3 Objectives of the Project

The main objectives of the project are to designed and develop an Intelligent Automatic Robotics Manipulator for landmine Detection and Retrieval using embedded and Robotics Technologies and the specific objectives adopt for the robot is following:

- To design and implement a mobile robotics platform that controlled by an ESP32 microcontroller as well as ESP32-S3 camera module for live monitoring.
- To integrate a metal detection sensor in the robot for identification of metallic objects and surface-level metallic object detection.
- To implement a GPS-based navigation for scanning of the selected area which define the navigation system for scanning.
- To develop a robotics arm which is capable of precise and stable piking of detected objects.
- To enable autonomous operation with minimum hazard human and work without human interference.
- To create a cost-effective and suitable prototype of a robot for future enhancement which work more smooth with advance technologies.

1.4 Scope of the project

This project's scope includes the design, development and implementation of Intelligent Automatic Robotics Manipulator in a war are for landmines Detection and this study covers the following aspects of the intelligent robotics field:

1.4.1 Mechanical and Electrical Design

Design and integration of a robotics manipulator and the mobile platform for the movement of the robot is developed by using ESP32 microcontroller, motor driver and different sensors for object detection and multiple operations.

1.4.2 Sensing and Detection Systems

Implementation of metal detection and object sensing algorithm and techniques used to identify potential landmine-like objects detection with high accuracy in different environment.

1.4.3 Control and Navigation

The development of the robot is mainly based on ESP32 microcontroller and there is a control algorithm for its movement and obstacle avoidance for better manipulation operation.

1.4.4 Human-Robot Interaction

Basic user interface of the web server for monitoring and manual control of the robot make it easier for user to interact with the robot and control it easily.

1.4.5 Experimental Validation

Testing of the robot in a controlled environment is very important and it is done by using a landmine-like object and evaluate the detection accuracy in its result and system reliability when

performing action. Although the primary focus in the result evaluation of the project is the detection of landmine and the design can be adapted for similar hazardous environments for object detection and handling that require automated inspection and service operations.

1.5 Application

The Intelligent Automatic Robotics Manipulator has wide practical application in hazardous and security-sensitive environments and the primary use of the robot is in humanitarian demining operations where it can detect and safely remove landmines without making humans into danger(Malizia et al. (2026). The system can be developed in post-conflict regions to clear agricultural land and residential area helping communities return to normal life.

(Harikrishna et al. (2025) It is also useful for military and defense operation for battlefield clearance and surveillance and the robotics system can be applied in border for security purpose and in restricted zones for automated inspection and threat detection. Additionally, it can be used in industrial are for safety and identification and handling of hazardous metallic objects in mining and construction sites. The project also serves as a research platform for college students in the field of intelligent robotics, embedded systems and autonomous navigation studies which make future development in smart robotics technologies.

1.6 Significance of the study

The deployment of intelligent Automatic Robotics Manipulator is expected to provide numerous advantages:

- Reduce risk to human life by eliminating direct human involvement in landmine detection and handling.

- This study provides a safe system and alternative to traditional manual searching method.
- This improves the speed of detection and efficiency of hazardous area scanning and clearance.
- This study also enhances accuracy and consistency of robot in detecting metallic threats and also enable precise GPS-based mapping of dangerous zones.
- It also demonstrates the practical integration of embedded robotics system and mechanical manipulation of landmine like object in the ground.
- This study also contributes in the development of autonomous robot and intelligent robotic systems
- Serves used for the movement of arm act as a foundation of future research in smart and safety-critical robotics.

Additionally, this study adds to the broader field of service robotics by showcasing the application of autonomous Robots for landmine detection and manipulation thereby broadening the potential focus on motion planning and manipulation of robot.

1.7 Technological Framework

The Robotics Manipulator utilize multiple modern technologies to perform smooth detection and manipulation operations and these technologies work together to for detection, navigation, manipulation and monitoring of environment with hazardous objects. The main technological components for detection used in the robot are as follows:

- **Embedded System:** An ESP32 microcontroller is used as the main controller in the robot which manage sensors, robotics arm movement and communication between different hardware components and also make send and receive data from another ESP32 cam module.

- **Computer Vision:** A ESP32-S3 Camera module is used in the robot for its vision system to monitor the environment and observe the surrounding environment
- **Metal Detection Sensor:** A metal detector is used to detect metallic object under the ground which may indicate the presence of landmines or unexploded metallic devices can be detected using this sensor.
- **GPS Navigation:** A GPS module is used to determine the robot's location and assist in scanning and mapping the selected area for detection.
- **Robotics Manipulator:** A mechanical metallic robotics Arm with 6-DOF servo is used and integrated with the mobile platform for carefully pick or handle of detected objects like landmine or metal object without direct human involvement.
- **Ultrasound Sensor:** An ultrasonic sensor is used for obstacle avoidance which enable the robot work more efficiently without getting accident.
- **Robotics Tank:** A Tank Chassis Caterpillar T200 used. The robotics arm and GPS is mounted on the top of the tank, Metal Detector Sensors are on the front of the tank so that can detect metallic object and quickly send signal to the microcontroller.
- **Web Interface:** A website interface is developed by using HTML and CSS have both manual and automatic option to operate the robot. There are two modes on the interface manual and automatic enable using a button.

These technologies collectively form the technological backbone of the proposed robot system, enabling safer landmine detection, autonomous operation as well as manual control and reduce risk to human in hazardous environments.

1.8 Social and Ethical Perspectives

Like every robotics system, it is important to consider the social and ethical aspects of this project as well. The proposed robotics manipulator for landmine detection is designed to protect human life by reducing the need for direct human involvement in dangerous area. The system should operate responsible and safely without causing harm to people and infrastructure. The robot must follow safety standards and operate with reliability and transparency and the public acceptance and trust in robotics systems and harmless operation of the robot are also important for the successful use of robotics technology. Proper safety measures, responsible use of data and careful handling of hazardous object are essential to ensure that the system contributes positively to society while maintaining ethical responsibility.

CHAPTER 2

Literature Review

2.1 Introduction

Landmines have first been used in the American Civil War in the 1800s (Singh Panwar et al. (2024). Landmines are a persistent global threat, buried across many regions and causing civilian and soldier's injuries, deaths and socioeconomic disruption. Traditional landmine detection method are slow, dangerous, risk of human life and costly motivating the development of robotics system that can detect and interact with buried explosive devices that can safely identify the landmine and pick from ground while keeping human at safe distance.

2.2 Challenges in Landmine Detection

Detecting landmines is inherently difficult due to variable soil conditions, heterogeneous environments and the small size and low signature of many miens and these issues often lead to high false positive rates in detection systems and also reduce accuracy and performance(Gallagher et al. (n.d.). For Example, soil with high and strong magnetic property and metal-like property specially in tropical and high mineral areas can interfere and make disturbance in the sensor signal and this disturbance make it harder for sensor to detect the landmine accurately and lower the sensitivity of sensing. This problem is particularly acute for modern anti-personnel mines that contain minimal metal which forces detectors to operate at high sensitivity yet still struggle to discriminate actual threats from soil effects or scattered scrap metal, further slowing detection and reducing operational effectiveness. Studies have shown that the physical and electrical characteristics of different soils directly influence detection performance and misclassification

rates, making consistent identification across diverse terrains one of the biggest challenges in landmine clearance.

2.3 Conventional and Manual Demining Techniques

(Gonzalez De Santos et al. (2005) Manually checking the ground by hand for landmine and trained animals (e.g., Dogs), and handheld detectors are common in history for landmine detection and clearance of the landmine mounted area. In manual condition, trained personnel carefully examine the ground by using rods or probs to locate buried landmines. similarly trained dogs are capable of detecting explosive materials through their strong sense of smell while handhold detectors are used to identify metallic components of landmines. However, these conventional methods expose operators to a high level of dangers as they must work very close to potentially explosive devices. In addition, these techniques are time-consuming, require extensive training and are often inefficient when dealing with large contaminated areas. (Gao et al., 2025)Environment conditions such as soil type, vegetation and weather can also affect the accuracy and performance of these methods making landmine clearance a challenging and risk task.

2.4 Robotics Systems for Mine Detection

The Concept of Robotic Demining has been under investigation for several decades as researcher aim to reduce the risk associated with manual landmine detection. (Alsalem et al., 2026)Early work explored semi-autonomous robots capable of landmine affected areas and marking suspected landmine location. These systems helped reduce the direct exposure of human operators to dangerous environments and improved the safety of landmine clearance procedure. With the advancement in technology, robotics and embedded system, the modern research is now focusing on intelligent robotics systems that combine multiple sensors, autonomous navigation and

advanced control mechanisms. These systems are capable of detecting, locating and handling hazardous object with the minimal human intervention (Marcelo Saavedra et al. (2025). The integration of technologies such as computer vision, GPS navigation and metal detection sensors has further improved the efficiency, accuracy and safety of robotics landmine detection systems.

2.5 Mobile Platforms and UGVs

The Unmanned Ground Vehicles (UGVs) give important mobility for covering large fie and area for landmine detection. Researchers have designed wheeled and track-based platform which equipped with advance navigation and sensing suites to systematically and efficiently search minefield. (Sineglazov & Lesohorskyi, 2025b)Wheeled platforms offer speed and energy efficiency on flat terrain, while tracked platforms provide better stability in rough environments.

2.6 Sensor Integration and Detection Technology

Metal detectors remain the fundamental sensors for identifying metallic components of landmines. To improve detection accuracy and reliability of the robot, many researchers integrate different prototype of metal detectors with additional sensor types such as ultrasonic sensors, IR sensors and temperature sensors for advance detection. (Falcone et al., 2025)These multi-sensor approaches allow robotics systems to better distinguish the landmines from other metallic objects and detect partially buried or non-metallic mines and operate effectively under different environmental condition.

2.7 GPS and IOT Integration

(Gallagher et al., n.d.-b)Modern robotics landmine detection systems are often developed with the help of GPS positioning with GSM or IOT based data links for accurately determine the landmine

location and correctly navigate in the selected area and transmit detection of results in real time. This enables the operators to monitor the robot remotely and analyze the operational of robot in environment and make strong decisions without being physically present or involve in hazardous area. IoT-enhanced designs further improve the connectivity, data logging and analysis capabilities of the robot allowing for advance mapping, predictive modeling and efficient reporting of cleared areas(2012 8th International Symposium on Mechatronics and Its Applications : 10th to 12th April 2012 : [Sharjah, UAE]), 2012).

2.8 Sensor Fusion and Multimodal Detection

(“Landmine Detection Robotic Vehicle Using ARM Cortex (STM32),” 2025) Sensor fusion takes data from metal detectors which is used to detect metals like landmine and imaging systems and other sensing techniques has been shown to reduce the false alarms and improve classification performance of the robot in mine detection.(Phule et al., 2026) Leveraging multiple sensors in the robot can help to differentiate multiple and different size of landmines.

2.9 Machine learning and AI Approaches

(Dwibedi et al., 2026) Recent advancements apply machine learning and computer vision algorithms to analyze sensor and image data from improved detection. For Example, researchers have developed IoT robotics that use real-time image classification to identify suspicious terrain features potentially indicating buried mines.

2.10 Data Sets and Algorithm Benchmarking

The datasets that is made for the landmine detection robot such as Mine Insight are created to help researchers in the robotics field and they provide different types of sensor data which include

image from different angles and light range which make it easier to test and improve algorithms. These datasets are crucial for training and validating AI systems for mine detection under varied conditions.

2.11 Manipulators and Robotics Arm in Demining

(Cobano et al., 2008)Robotics manipulators and arm are increasingly integrated into robotics demining to physically interact with or lift suspected objects without exposing humans to danger. These manipulators allow the robot to safely probe, pick or move potential landmines reducing the risk of accidental detonation by enabling precise handling of the objects and controlled manipulation of hazardous objects and the arms enhance the safety for human and reliability and effectiveness of landmine detection operations.

2.12 Navigation, Autonomy and Control System

Navigation of the robot in autonomous mode and control systems play a vital role in robotic motion planning and manipulation especially where need to complex task or landmine contaminated environments. (2017 *International Conference on Inventive Computing and Informatics (ICICI)*, 2017)Autonomous navigation and path-planning techniques allow robots to move safely through hazardous areas while identifying and accurately detect suspicious object or landmines. Autonomous or semi-autonomous systems also help reduce the workload of human operators while improving safety and operational efficiency. In addition to autonomous operation, the manual control system of the robotics manipulator gives a remotely control to the operators and user to control the robot and its manipulator arm when object is required and need to perform precise movement or careful handling of robot to deploy the robot. This hybrid approach combines

manual control with autonomous navigation control and provides greater flexibility and reliability to the robot and also safety during landmine detection and retrieval tasks.

2.13 Real-World Deployments and Limitations

Despite progress, many robotics mine detection system is still at the prototype or research stage. Limitations include cost, environmental constraints (e.g. deep burial, soil heterogeneity), and robustness in real minefields. Additionally, factors like battery life maintenance requirements and operator training also affect real-world adoption. Addressing these limitations are critical to making robotics demining systems both scalable and effective for long scale hazardous environment.

2.14 Summery

The review literature that landmine detection is a highly dangerous and technically challenging task when performed using conventional manual methods. Most studies highlight robotics and intelligent systems as effective alternatives for operating in hazardous environment while minimizing human risk. Mobile robotics platform equipped with metal detector, ultrasonic and imaging sensor using GPS/GMS communication have shown improved efficiency in locating potential mines. The integration of robotics manipulators enables safe inspection and interaction with suspicious objects, while the application accuracy and reduces false alarms. However, many existing solutions remain limited by high cost, partial autonomy, environmental sensitivity and insufficient real-world validation. Therefore, there is a clear need for a low-cost, modular and intelligent robotics manipulator, which directly supports the motivation and objectives of this study.

CHAPTER 3

Hardware Design and Implementation

3.1 System Overview

The proposed system is an autonomous / Manual mobile robot integrated with a robotics manipulator and multiple sensors for landmine detection. It is controlled by an embedded microcontroller that coordinate sensing, motion and manipulation.

3.2 System Architecture (Block Diagram)

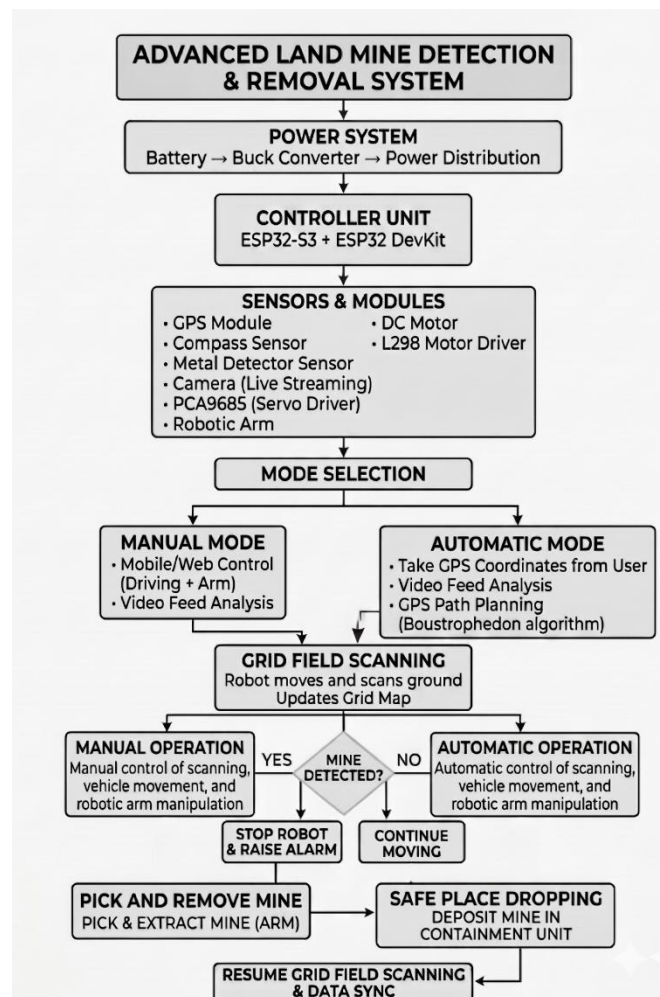


Fig. 3.1 Block Diagram

The ESP32 microcontroller is the brain of the project. It gets input from metal detector and ultrasonic sensors and processes the data then sends control signals to the motor driver and servo motor. This makes a remote monitoring and control system while the Battery source provides electrical energy to all components.

3.3 Hardware Design (Circuit Diagram)

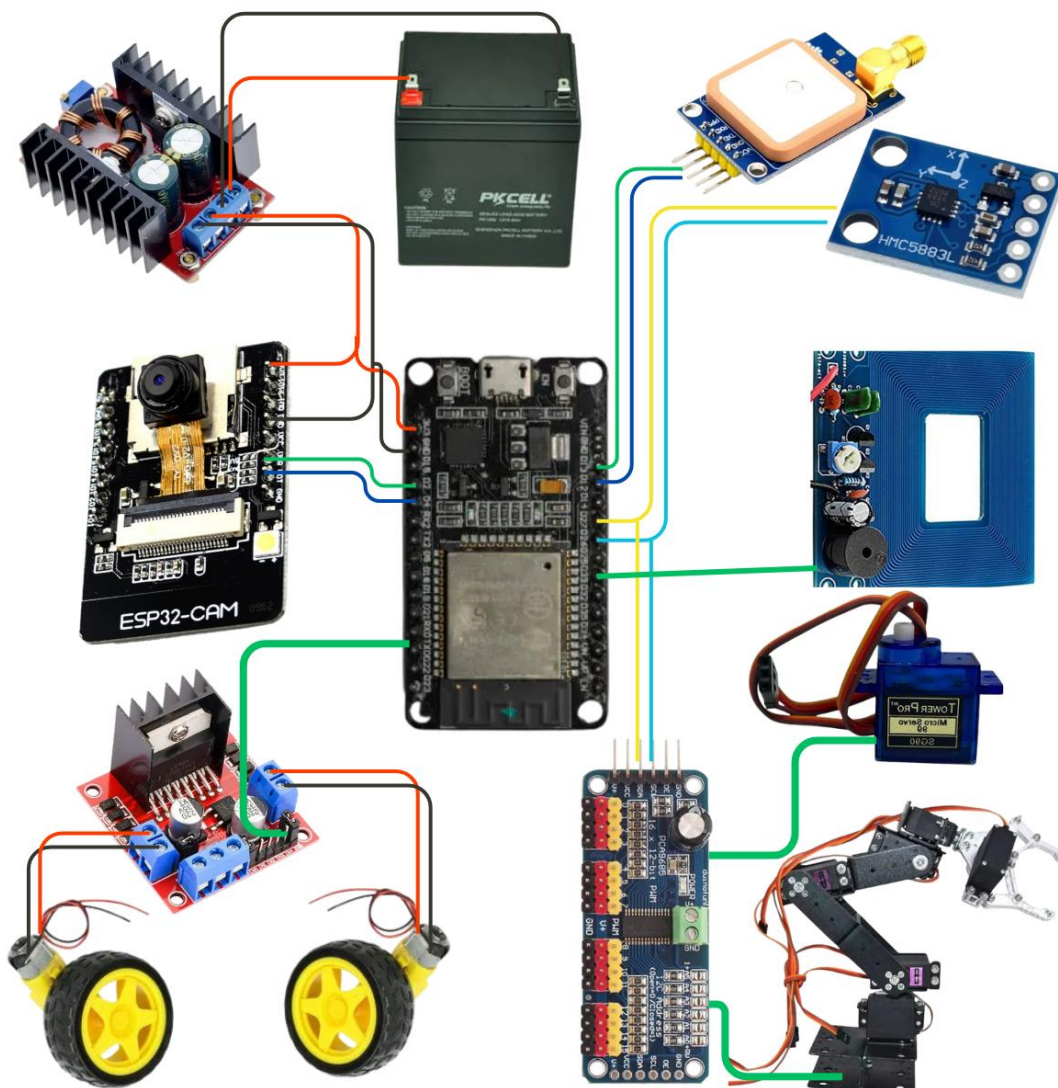


Fig. 3.2 Circuit Diagram

The circuit diagram Fig. 3.2 explains the overall wiring and circuit connection. The main controller is ESP32 and ESP32 cam is used for real-time live monitoring. Buck converter is used to divide the large current into a small and required current and the L298N motor driver is used to control the DC motors that move the robot. A metal detection is on the front of the robot with a servo continuously scanning the area and once detect a metallic object, send signal to the ESP32 and the Robotics Arm will activate and manipulate the landmine from land and move on for further scanning. The PCA board is used for the 6-DOF Servos used in the robotics arm. It has a 16 pin connection capability that enhance the arms performance.

3.4 Microcontroller and Processing Unit

The robotics manipulator hardware is design by using ESP32-WROOM module as well as ESP32-S3 CAM module, responsible for controlling all hardware modules, processing sensor data and managing communication between different system components. Both modules are based on ESP32 chip and provide powerful processing capability, wireless connectivity and efficient power management suitable for embedded and IoT applications.

3.4.1 ESP32- Devkit

The ESP32-Devkit module serves as the main controller of the system. It is a low power; high performance microcontroller developed for different IoT applications. The module is built around a dual-core 32-bit Xtensa LX6 Processor which can operate at a clock frequency of up to 240 MHz providing sufficient computational power for real-time processing task.



Fig. 3.3 ESP32 Devkit Module

The ESP32 has following features and functionalities

- ESP32 is a Dual Core 32-bit Microcontroller
- Its Operating frequency up to 24 MHz
- It has built-in Wi-Fi and Bluetooth module
- Multiple GPIO pins with ADC, DAC PWM support
- UART, SPI and I^2C communication interfaces
- It has Integrated timers, consume low-power and deep-sleep modes
- It has On-chip memory and flash support.

3.4.2 ESP 32 –S3 Cam Module

The ESP32-CAM module is a compact microcontroller board that integrates the ESP32 chip with an OV2540 camera sensor making it suitable for image capture and basic vision applications. This model also includes Wi-Fi and Bluetooth connectivity, enabling wireless image and video transmission and remote monitoring.



Fig. 3.4 ESP32-S3 Cam Module

The ESP32-Cam contains:

- 32-bit ESP32 processor
- OV2640 2-megapixel camera
- MicroSD card slot for image storage
- Wi-Fi and Bluetooth connectivity
- Multi GPIO pins for external interfacing

In the robotics manipulator, ESP32-Cam is mainly used for live and remote monitoring and visual data processing. Data can be transmitted through Wi-Fi for monitoring the location of the robot and the status of the area where the robot operating. The microSD card interface allows local storage of data if required.

The ESP32-Cam work together with ESP32-WROOM module where;

- The ESP32-CAM handle the visual sensing tasks by sending the data to web interface
- The ESP32-WROOM module performs main system control and data processing

3.5 Sensors and Actuators

Sensors and actuator as essential components of the robotics manipulator. The landmine detection robotics manipulator is fully mounted with different sensor and actuator for smooth performance. These sensor and actuator work together when perceive signals from the area and perform certain action in effective and efficient way. The integration of sensors and actuators with the processing unit enables the system to monitor its surroundings, analyze the received data and respond accordingly. The ESP32-WROOM and ESP32-CAM modules process the information collected by sensors and generate control signal to operate the actuators.

3.5.1 Metal Detector

A metal detector is use which operates by generating an electromagnetic field to detect landmine and when a metallic object come under the area, it detects and send signals which converted into an electrical output signal.



Fig. 3.5 Metal Detector Sensor

It is the primary sensor used to detect metallic components of buried landmine like objects. The main important features are;

- Detects metallic objects beneath the surface.
- Works using electromagnetic field sensing.
- Provide quick and reliable detection response.
- Can identify low and high metal content objects.
- Easy to interface with microcontroller (ESP32/Arduino).

3.5.2 Camera

The camera module used is the ESP32-CAM which includes an OV2640 2-megapixel camera sensor for visual task. It is capable of visual recording, image capture and live streaming and transmit the data through Wi-Fi for monitoring and analysis. The camera is used to visually observe the environment and assist in detecting objects or landmine in the operational area. The ESP32-CAM processes the area and can store them on a microSD card or send them to a remote system. It sends the data to the web interface and a user can live monitor the operation of the manipulator.



Fig. 3.6 ESP32-CAM

3.5.3 Buck Converter

A buck converter is a DC-DC converter used to step down the 12V battery voltage to lower levels (5V/3V) required by the components like ESP32 and sensors in the system. It works by switching

and regulating voltage efficiently, ensuring stable power supply and protecting components from damage.

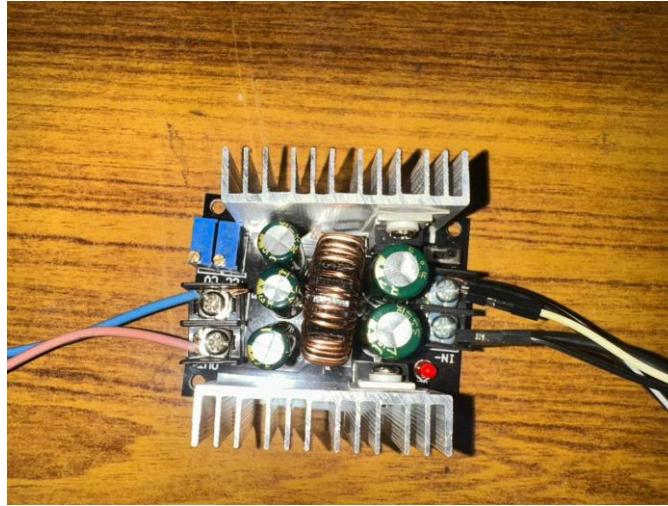


Fig. 3.7 Buck Converter

3.5.4 DC Motors

4 DC motors of 12V are used that convert energy into mechanical motion. This 4 DC motor used for wheel of the car to move in the area. Their speed and direction are controlled by varying the voltage and magnitude through a motor driver.

They provide movement to the mobile robotics platform mean the car which move in the certain are for scanning and it enable area coverage for landmine detection.



Fig. 3.8 DC motors for Wheel

3.5.5 Motor Driver (L298)

A Motor driver used to control actuator which provide connection between interface, the microcontroller and the DC motors. It amplifies control signals and allows bidirectional motor rotation. It enables safe and efficient control of the robot movement without overloading the microcontroller.

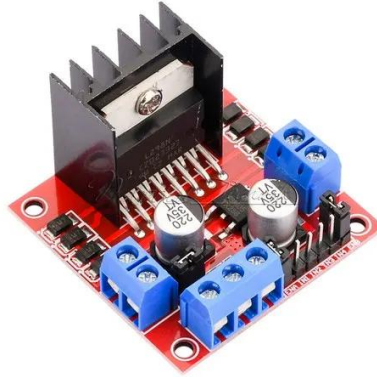


Fig 3.9 L298 Motor Driver

3.5.6 Servo Motors

The Servo motors is used for the movement of manipulator (Arm). It is 6 DOF and operate using pulse width Modulation (PWM) signal to achieve precise positioning They control the joint of the robotics manipulator, allowing accurate movement for inspection and handling of detected objects.



Fig. 3.10 Servo Motor

3.5.7 GPS Module

The GPS module is part of the manipulator that finds landmines. This thing is really helpful because it can track its location all the time and give us the position. The system can record the latitude and longitude of the landmines it finds which's really useful for making maps of areas that have landmines. This makes it safer, for people who are working in those areas. You can use the GPS information to watch what is happening live on a website. You can also use it to see what happened after the mission is over. The GPS module makes it easier to find all the landmines. It does this by reducing the chance that some areas will be missed or that the robotic manipulator will have to go to the same area again. In the future we might be able to use better GPS and new mapping techniques to make the robotic manipulator that finds landmines work even better.



Fig 3.11 GPS Module

3.5.8 PCA Board

The PCA9685 board is 16 channel and 12-bit motor controller that precisely control the 309 movement of the motors (PWM motors). we can control 16 servos using this PCA board at a time and it send and receive signal through I2C protocol. This board play a vital role in controlling the servo motors of the 6-DOF robotics arm's savor motors.

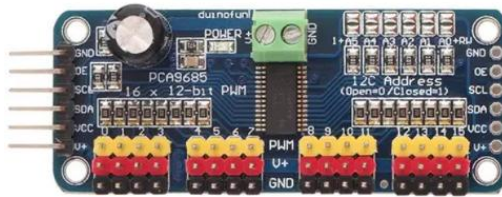


Fig 3.12 PCA Board

3.5.9 QMC5883L Compass

The QMC5883L Compass is a 3-axis compass that is use for the movement for automatic 313 robot to move freely in the area and we use this module for the effective and accurate 314 movement and navigation of the robot.



Fig 3.13 Campas

3.5.10 Robotics Manipulator

A multi-DOF (6 DOF) robotics arm driven by servo motors is used to inspect and handle detected objects without human contract. The Arm is metallic and can carry 1 kilogram weight. When the

metal detector detect object the microcontroller send signal to the arm based on the signal received from metal detector and the arm pick the landmine or the metal object from ground.

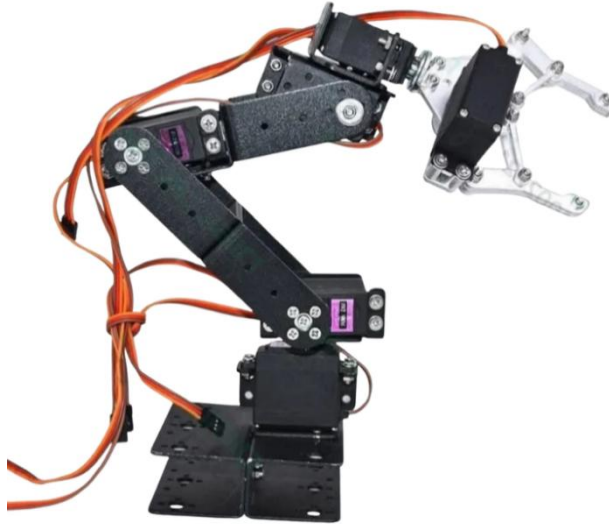


Fig. 3.14 Robotics Manipulator

The Manipulator has following important features and functionalities;

- 6-DOF (base rotation, shoulder, elbow, wrist pitch, wrist yaw and wrist roll)
- High positional and orientation flexibility
- Servo-motor based precise joint control
- Capable of complex 3D movement
- Programable motion and repeatable accuracy
- Grasping and lifting detected objects safely
- Precise positioning near suspicious landmine-like object
- Remote and automated operation through ESP32
- Minimize human involvement in hazardous handling

3.5.11 Mobile Robot Platform

A Wheeled chassis with 4 DC motors is used to move the robot over the test area. This body is made of plastic material. The motors is connected to the motor driver which enable the motors to move in the area while providing necessary power. The slow and controlled movement ensure accurate sensing and operational safety

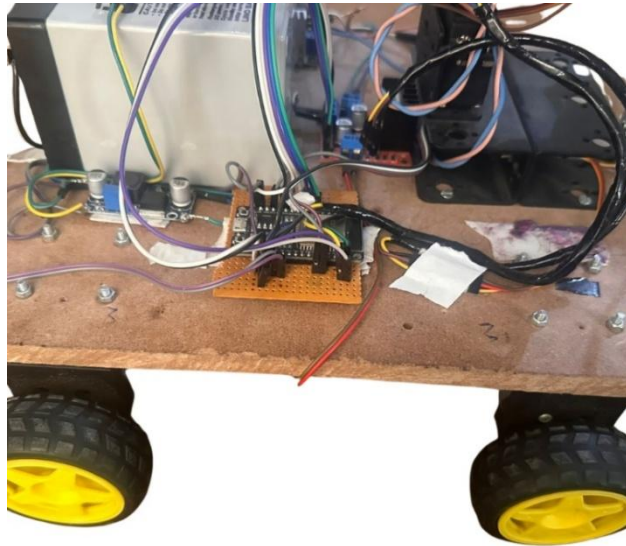


Fig. 3.15 Mobile Robt Platform

3.6 Power Supply and Distribution System

A 12V 7A Lead-Acid powerful rechargeable DC Battery is use to provide proper power supply each component providing a total energy capacity of 84Wh and make the robot's working long-lasting. It is rechargeable, once its charge low, we can easy charge it and use again instead of buying new one. It makes the robot more autonomous and professional. The battery is selected due to its adequate voltage capacity, long operating life, rechargeable ability and ability and power distribution architecture utilizing DC-to-Dc Buck Converter.



Fig. 3.16 Power Supply (Battery)

Since different components in the system require different voltage levels, proper voltage regulation and power management techniques are implemented to ensure safe and efficient operation. The Important factors in power distribution system include following:

3.6.1 Voltage Regulation Module

The system utilizes two primary regulation stages to ensure the stability and prevent interference between the high-torque actuators and sensitive logic electronics:

- **High-Current Actuator Supply:** A 300W 20A adjustable DC-DC Buck Converter is dedicated to the robotics arm and the scanning assembly and the output is regulated with 6V power. This module was selected to handle the high current demands of the six MG996R servo motors used in the Robotics Arm. Since each MG996R can draw a peak stall current of 2.5A, the converter is configured with 10A current limit to provide a balance between high-torque performance and circuit protection.
- **Logic and Sensor Supply:** Two HW-411a (LM2598) Buck Converters are used to provide an isolated power for the logic module and Auxiliary Actuation.

The logic module regulated to 5V to power the ESP32 microcontroller and the metal detector. Isolation from the motor rails ensures that back-EMF and switching noise from the servos do not cause system resets or sensor inaccuracies. While the Auxiliary Actuation is a secondary module provides 6V specifically for the scan servo to maintain consistent scanning speeds independent of the robotics arms movement.

3.6.2 Power Requirements Summary

Table 3.1 Power requirement

Subsystem	Operating Voltage	Primary Regulation	Max Current Limit
Robotics Arm (6xServos)	6 V	300W 20A Buck	10.0 A
Control Logic (ESPs)	5 V	LM2596 (HW-411a)	2.0 A
Metal Detector	5 V	LM2596 (HW-411a)	2.0 A
Scanning Servo	6 V	LM2596 (HW-411a)	2.5 A

3.6.3 Battery Management System (BMS)

A good battery management system ensures the safe and efficient operation. The BMS protect the battery from conditions such as overcharging, deep discharge, overheating and short circuit. By maintain the battery with safe operating limit, the BMS improve the safety, reliability and life of the power system.

3.6.4 Charging and Backup

The battery used in the robotics manipulator can be recharge using power adapter. When the battery level drops below a certain limit, the system can be powered down or recharged to maintain proper operation. Rechargeable battery systems allow the robot to be used repeatedly without replacing the power source, making the system more economical and practical for long-term use.

3.7 Software Integration and Control Algorithm

The ESP32 devkit and ESP32-S3 cam module both are programmed with Arduino IDE software to read sensor inputs from environment and send the data from one ESP to another through UART Communication protocol at a baud rate of 115,200 and run the logic continuously and make decisions autonomously. It gives command to the robot for movement of actuators using GPIO pins 19 and 21, regularly checks the ultrasonic sensors for obstacles and keeps detecting values from the metal detector for objects that inside the ground like landmines. Additionally, the logic produces Pulse Width Modulation (PWM) signals to precisely find the position and move manipulator's joint and provide appropriate speed for DC motors. It also manages the signal from GPS sensor which help the robot to decide where to work and where to not operator.

3.8 Mechanical Structure

Operating: The chassis of the robot platform provide stability and alignment by supporting all mechanical and electronic components of the robot. It offers a stiff platform on which the robotic manipulator, microcontroller, sensors, and motors can be mounted and the accuracy of operation of a sensors and actuators of robot is ensured by the ability of mechanical structure of robot that absorb small shocks and vibrations during each movement. Additionally, it keeps the manipulator

stable from falling over when it is extended by maintaining appropriate weight distribution and also to enable smooth and controlled motion throughout the testing environment, wheels and motor are integrated into the chassis.

Purpose: The purpose in the project is to maintain structure and facilitate easy movement throughout the testing area by keeping all parts securely in place and provide safe operation capability to the robot in the required area.

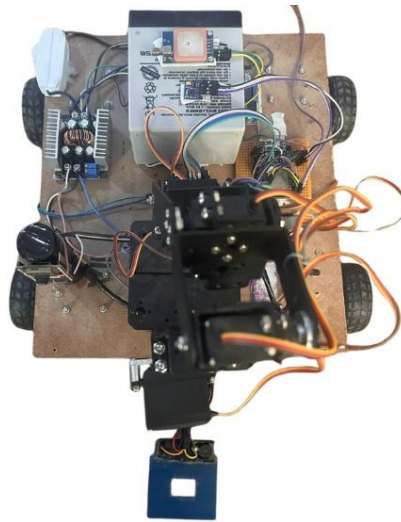


Fig. 3.17 Mechanical Structure

3.9 Actuator and Motor Specification

This section details the motion control components including their specific operational ranges and electrical characteristics.

3.9.1 Servo Motor Mapping (Angular Constraints)

To prevent the mechanical interference and protect the structural integrity of the robotics arm and software defined angular limits have been implemented. The system utilizes 6 MG996R high-torque servos for the arm and one MG90S for the scanner.

Table 3.2 Servo Motor Angles

Actuator	Component	Operational Range
Servo 0	Griper	80° – 150°
Servo 1		0° – 150°
Servo 2		0° – 120°
Servo 3		0° – 100°
Servo 4		0° – 120°
Servo 5	Base Rotation	40° – 150°
Scanner	MG90S (Metal Detector)	0° – 130°

3.9.2 Drive System and Current Analysis

The locomotion of the project is managed by a dual-motor drive system controlled via pulse width modulation (PWM).

- L298N Motor Driver: A dual H-Bridge Driver is utilized to control the 12 V motors and it supports a continuous current of 2A per channel, providing sufficient overhead for the geared motors.
- 12V DC Geared Motors: These motors provide the necessary torque for the chassis and under normal load, each motor draws approximately 300 mA to 600 mA. In a stall condition (High Resistance), the current can peak at 1.5 A which remains within the safe operating limits of the L298N driver.

3.9.3 Current Requirement Summary

Table 3.3 Current Requirement

Component	Operating Current (Avg)	Peak / Stall Current
MG996R Servos (x6)	500mA – 900mA (each)	2.5 A (each)
12V Gear Motors (x2)	500mA (each)	1.5 (each)
L298N Driver	-	4.0A total Capacity

3.10 Material and Development Cost

The main hardware elements in the creation of the robotic landmine detection system are shown in the following table along with an estimate of their market value. The suggested system is economically viable and appropriate for low-cost humanitarian applications, as this cost analysis shows.

Table. 3.4 Estimate Cost

Sr. No.	Component Name	Quantity	Unit Cost (PKR)	Total Cost (PKR)
1	ESP32-Devkit	1	2,000x1	2,000
2	ESP32-S3 CAM	1	1200x1	1200
3	Metal Detector Sensor	1	550x1	550
4	DC Gear Motors	4	1400x4	5600
5	Motor Driver (L298N)	1	500x1	500
6	6-DOF Robotic Arm	1	18,000x1	18,000
8	Chassis & Wheels	1	1700x1	1700

Sr. No.	Component Name	Quantity	Unit Cost (PKR)	Total Cost (PKR)
9	12V Meter	1	400x1	400
10	Power Supply & Batteries	1	2,500x1	2,500
11	12V Charger	1	1700x1	1700
12	Connecting Wires & Misc.	6	150x6	1500
13	Buck Convertor	2	400x2	800
14	Button	1	30x1	30
15	Servo for metal detector	1	300x1	300
16	Compass	1	2700x1	2700
17	PCA Board	1	4000x1	4000
18	Glue, Wiring etc	---	2000	2000
13	Shipment Charges	4000		
14	Total	49,480/-		

3.11 Summery

The design, integration, and operation of an intelligent robotic system for secure landmine detection are the main objectives of this project's methodology. The robot is designed using a moveable robotic platform like tank with wireless communication ability of ESP32-WROOM microcontroller with the help of GPS sensor. The Metal detection and ultrasonic sensors continuously detect and provide signal to the controller and a 6-DOF manipulator activate when detect some objects. ESP32-Cam module provides live monitoring of the operational area. Every

component is carefully selected to perform particular task and the manipulator enables accurate searching and handling ability of suspicious objects in the ground. The gear motors for wheel provide a controlled movement for the robot platform with the help of motor driver L298. In order to perform action on sensor data, navigate safely and react intelligently to threats or landmine, the system runs real-time control algorithms. While the power supply ensures continuous operation, the chassis and mechanical structure guarantee stability and dependability during movement. In order to ensure that the robotic system can safely and effectively detect, approach, and manipulate potential landmine objects in a controlled environment, the methodology generally places an emphasis on modular design, component coordination, and semi-autonomous operation

CHAPTER 4

Software Implementation

4.1 Introduction

The software of the Intelligent Vision-Based Manual/Automatic Robotics Manipulator developed using C++ programming language with the FreeRTOS framework for efficient real-time task management and system control. A web-based interface built with HTML, CSS and JavaScript enables user to interact with the system and control it in both manual and automatic mode. The integration of the landmine detection robot programming and web interface technology make the robot work smooth, reliable and responsive for operation of the manipulator. The software architecture of the robot is designed to handle multiple tasks and action such as sensor data processing, motor control and communication between modules and it also responsible for synchronization between the vision system and movement of a robot for accurate operation. Furthermore, the working fundamentals of the robotics manipulator is optimized for performance and scalable for future enhancement and improvement for advance features.

4.2 Design Principles

The following design principles are applied in the Land mine Detection Robotics Manipulator:

- **Modularity:** The system is divided into separate module such as vision processing, control and communication. This improves development, testing and maintenance.
- **Scalability:** The design of the robot allows us for addition of new features or hardware components in the system with minimum changes and supporting future expansion.

- **Reliability** The robotics system is designed to handle machine errors and work continuous operation providing a stable performance in their action during different task.
- **Real-Time Performance:** Using FreeRTOS we make the robot more powerful that the tasks like sensor processing, motor control and communication between each ESP32 module are executed within defined time constraints.
- **User-Friendly Interface:** The web server interface which is designed by using HTML, CSS and JavaScript is to be very simple and attractive that make the robot easy to control in manual mode for user and monitor the system easily.

4.3 FreeRTOS Framework

The software architecture of the Intelligent Vision-Based Manual/Automatic Robotics Manipulator is designed with C++ programming language by using FreeRTOS that is a framework of C++ and it is a lightweight real-time operating system which is widely used in embedded systems. FreeRTOS provide efficient task scheduling, timing control and resource management which make it suitable for handling multiple operations simultaneously in the ESP32-Based Robotics Systems. The multitasking capability of FreeRTOS make it critical for operations that are executed within defined time constraints, enhancing real-time performance and additionally, FreeRTOS support inter-task communication and synchronization mechanisms of programming that allow smooth data exchange between different robotics components. It has an efficient memory for usage as it supports multi-tasking action so its flexibility makes it ideal for embedded system and robotics and IoT applications which making it reliable and scalable of performance for the manipulator system in the landmine detection Robot system

4.4 Software Functional Block

The software architecture of the Landmine Detection Robotics Manipulator includes several important parts, in which each part plays specific function:

4.4.1 Navigation Module

This module enables the Intelligent Vision-Based Manual/Automatic Robotics Manipulator to move within its environment and perform tasks such as path and motion planning, localization and motion the car. By using data from the camera and sensors, the system analyzes its surroundings and make decisions for safe efficient navigation. The GPS navigation logic is implemented by using control and real-time processing ability of ESP32 which allow the robot to adapt to different conditions and perform scanning in the given coordinate.

4.4.2 Perception Module

The perception module in the robot process visual data from the camera of ESP32-S3 module to understand the environment and send the data to liver sever where user can monitor easily on the web interface and it performs tasks such abject detection and live streaming through web server which are essential for manipulation and navigation for the robot to work smooth in the given area.

4.4.3 Control Module

The control module manages the overall operation of the Intelligent Vision-Based Manual/Automatic Robotics Manipulator and it controls the movement of motors in arm of the robot, GPS for navigation, metal detector and actuators based on commands and input from user which received through the web interface and decisions from the navigation module. It make sure the robot to work properly in coordination between manual and automatic modes by smooth

switching from the sever and accurate execution of task. The control logic implemented using FreeRTOS enables real-time response and synchronized operation of all the system component.

4.4.4 Manual Control

The robotics manipulator is based on dual mode control algorithm and the robot can operate manually from the web interface developed by HTML, CSS and JavaScript. User can select the mode either go with manual or automatic. In the manual section the robot can control from the interface with the help of mobile phone or a laptop. ESP32 read signal from the interface and send it to the actuators as per the signal receive from user and user can monitor the environment with the help of ESP32 can module which communicate with the ESP32 Devkit with UART communication protocol. In this mode, user personally and by hand handle the robot and its working and operate the car, robotics arms and the scanner. To operate each part, there a respective button on the interface and the robot perform acting according to the user input form the button.

4.4.5 Autonomous Path Following.

Another mode is Autonomous and the robotics manipulator work autonomously by navigating GPS and Campas sensor. The following steps involve in the autonomous operation of the Robot.

Step 1: Generating the Map (Linear Interpolation): To create the zig-zag scanning pattern, the robot calculates exact internal waypoints between your boundaries (G_1, G_2, G_3, G_4). It uses Linear Interpolation (LERP). Given a number of lanes (N=3), the robot calculates a fractional step for the current lane (i): $\text{Fraction} = \frac{i}{N - 1}$. Then, it calculates the left and right boundary coordinates for that specific lane. For the Left Edge (interpolating between G_1 and G_2): $\text{Lat_left} = (\text{Lat_G1} + \text{Lat_G2} - \text{Lat_G1}) \times \text{Fraction}$ $\text{Lon_left} = \text{Lon_G1} + (\text{Lon_G2} - \text{Lon_G1}) \times \text{Fraction}$. For the Right Edge (interpolating between G_4 and G_3):

$$\text{Lat_right} = \text{Lat_G4} + (\text{Lat_G3} - \text{Lat_G4}) \times \text{Fraction}$$

$$\text{Lon_right} = \text{Lon_G4} + (\text{Lon_G3} - \text{Lon_G4}) \times \text{Fraction}$$
 By connecting these left and right points in an alternating sequence, it creates the zig-zag path arrays.

Step 2: Distance to Waypoint (The Haversine Formula): To know if the robot has arrived at its target (distance ≤ 1.5 meters), it must calculate the shortest distance over the Earth's curved surface. This is the math inside `gps.distanceBetween()`.

Let the robot's current position be (ϕ_1, λ_1) and the target waypoint be (ϕ_2, λ_2) , where ϕ is latitude and λ is longitude (converted to radians). First, calculate the differences: $\Delta\phi = \phi_2 - \phi_1$ $\Delta\lambda = \lambda_2 - \lambda_1$ Apply the Haversine formula: $a = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos(\phi_1)\cos(\phi_2)\sin^2\left(\frac{\Delta\lambda}{2}\right)$ $c = 2 \cdot \arctan2(\sqrt{a}, \sqrt{1-a})$ $\text{Distance} = R \cdot c$ (Where R is the Earth's radius, approximately 6,371,000 meters).

Step 3: Target Bearing (Forward Azimuth calculation): To figure out which compass angle the robot needs to face to reach the next waypoint, it calculates the Forward Azimuth. This is the math inside `gps.courseTo()`. Using the same latitudes (ϕ) and longitudes (λ) in radians: $X = \cos(\phi_2) \cdot \sin(\Delta\lambda)$ $Y = \cos(\phi_1) \cdot \sin(\phi_2) - \sin(\phi_1) \cdot \cos(\phi_2) \cdot \cos(\Delta\lambda)$ Calculate the angle θ in radians: $\theta = \arctan2(X, Y)$ Convert θ from radians to degrees and normalize it to a standard 0° to 360° compass heading: $\text{Bearing} = \left(\theta \times \frac{180}{\pi} + 360\right) \bmod 360$

Step 4: Steering Error & Shortest Turn (Modular Arithmetic): Now the robot has two numbers: its Current Heading (from the QMC5883L compass) and the Target Bearing (from the math in Step 3). It calculates the raw difference (Error) $\text{Error} = \text{Target Bearing} - \text{Current Heading}$

Because a compass is a circle, an error of 350° actually just means a 10° turn in the opposite direction. To prevent the robot from doing a massive, unnecessary spin, it uses modular arithmetic to find the shortest turn path:

- If $\text{Error} > 180^\circ$ Right arrow $\text{Error} = \text{Error} - 360^\circ$
- If $\text{Error} < -180^\circ$ Right arrow $\text{Error} = \text{Error} + 360^\circ$

The Steering Decision: The code checks the absolute value of this normalized error: $|\text{Error}|$.

- If $|\text{Error}| \leq 10^\circ$, it drives straight ahead.
- If $|\text{Error}| > 10^\circ$ and $\text{Error} > 0$, it spins the right track faster to turn Right.
- If $|\text{Error}| > 10^\circ$ and $\text{Error} < 0$, it spins the left track faster to turn Left.

4.4.6 Web Interface

The web interface is developed by HTML, CSS and JavaScript. It includes a user-friendly interface with easy-to-use buttons. Initially the ESP32 can generate IP Address and with the help of this IP user can control the robot from the interface. Once the user successfully get the IP address and go to any browser, the interface will open and there are two buttons (Manual and Automatic).

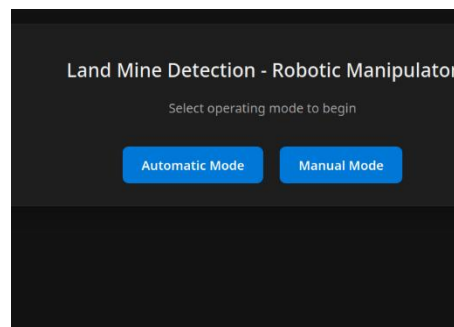


Fig 4.1 web interface

If the user selects manual then the user need give every command to control, detect and safely manipulate the landmine. User need to control each 6 servo of the robotics arm to safely manipulate the landmine

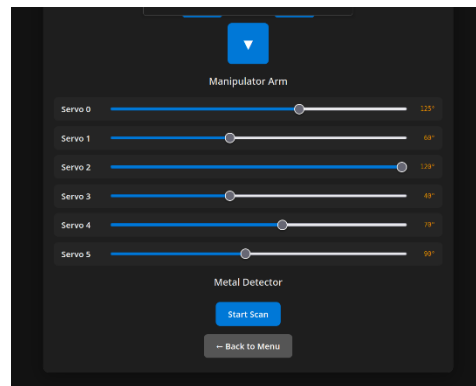


Fig 4.2 web interface Manual Mode

While if the user selects autonomous mode, then the interface go to autonomous mode and it has user input section for coordinate. User need to input 4 coordinates of latitude and longitude for GPS. After this a start button is on the last of the interface and once click on start the robot start scanning in this selected coordinate and if a landmine object detected, the Robotics Am activated and manipulate the object and start scanning again until the selected area completely scanned. All these actions performed the robot autonomously.

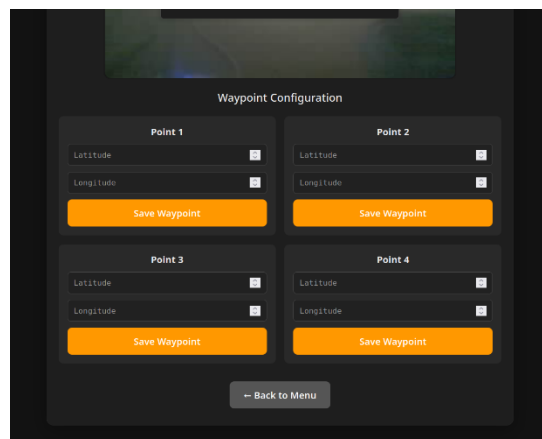


Fig 4.3 web interface Automatic Mode

In the both mode (Manual and Autonomous) a live streaming will be show on the interface in both mode and this happened with the help of ESP32 cam which continuously stream the environment and send the signal to the interface where user can see the environment.

4.5 Integration and Communication

All modules of the landmine Detection Robotics Manipulator are integrated through a structured communication system based on the FreeRTOS framework and Wi-Fi connectivity between ESP32 and ESP32-S3 Camera Module. Each module such as control, vision and navigation operates as an independent task and communicates with others through shared data and synchronization mechanisms. This allows efficient data exchange through UART communication between both ESP32 within the system. For example, the vision module process camera and the data to web server for live streaming. The metal detection sends data to ESP32 to control modules while the web interface transmits user command to the controller via Wi-Fi. After that he manual control module coordinate actions such as motor movement of arm and task execution. As the proper communication links are mention in the system, this modular communication approach add new features in the robot and the modules has a minimum impact on the existing robot system. Additionally, the distribution of the processing of task across multi ESP32 units improves robot system performance and reliability.

4.6 Implementation

The implementation process of the robot (Intelligent Vision-Based Manual/Automatic Robotics Manipulator) software include the following technologies and tools:

- **Programming Language:** C++ programming language is used for the entire system programming and real-time control using FreeRTOS framework and HTML, CSS and JavaScript are used to develop the web-interface for user interaction and manual control.
- **Framework:** FreeRTOS is a framework that are used as the real-time operating system to manage the multiple tasks of the robot such as sensor processing, motor control and communication efficiently.
- **Communication:** Inbuilt Wi-Fi communication of ESP32 is use between ESP32-S3 Camera module for web server Interface control and UART communication is formed between ES32 devkit and ESP32-S3 camera module which is for data exchange between camera unit, control unit and web server interface.
- **Library and Tools:** Various embedded and web development libraries are used to handle hardware interface, image processing and web server functionality, which speeds up development and improve system performance.
- **Testing and Validation:** The system us tested in real hardware conditions through multiple trials to validate functionality, performance and reliability before final deployment.

4.7 Summary

The software implementation of the Landmine Detection Robotics Manipulator is designed to be modular, scalable and efficient by utilizing the capabilities of the FreeRTOS framework and embedded system programming. This architecture enables smooth integration of different modules such and control, navigation, vision and detection ensuring reliable communication and real-time performance. As a result, the system successfully supports both manual and autonomous operations, allowing accurate manipulation, efficient navigation and effective user interaction as intended.

CHAPTER 5

Testing and Performance Evaluation

5.1 Introduction

This chapter presents the testing and performance evaluation of the Intelligent Vision-Based Manual / Automatic Robotics Manipulator for Landmine Detection. The main objective is to demonstrate that the proposed system can reliably and efficiently perform landmine detection and manipulation tasks in both controlled and real-world environments. This chapter describes the testing procedure adopted, including both manual and autonomous operation modes, along with the evaluation criteria used to measure system performance. Key performance indicators such as detection accuracy, response time, manipulator precision and system reliability are analyzed. The results obtained from various experiments are discussed to validate the effectiveness, safety and robustness of the proposed robotics manipulator.

5.2 Purpose of Evaluation

The main purpose of this designed structure evaluation is to validate the effectiveness and reliability of the Landmine Detection Robotics Manipulator in real-time operational environments. The evaluation purpose is to examine whether the robotics system can accurately detect landmine-like objects or not and safely manipulate them under different conditions. The robot was tested to determine or examine its ability to handle landmine and real-world challenges such as different environmental conditions, variations in lighting, presence of noise in visual data and obstacles in the environment. Further, to examine the performance of robot in both manual and autonomous

modes to ensure smooth operation, precise control of the manipulator and safe interaction with hazardous objects.

5.3 Importance of Testing

Testing plays a crucial role in any system design and also it is important for the development and deployment of robotics manipulator as it ensures that the system performs accurately, safely and reliably in real-world environments. Unlike traditional landmine detection and software systems, robot interact directly with the physical surroundings and make them more sensitive to uncertainties such as environmental variations, sensor noise and hardware limitations. In Robotics and Intelligent Systems, especially those based in vision and automation, testing is essential to validate the accuracy of perception, decision making of the designed robot and control mechanisms. It helps in identifying errors in object detections, misinterpretation of data and delay in response of manipulation which could lead to critical failures in practical applications.

5.4 Testing Procedures and Setup

The landmine detection robotics manipulator was evaluated and examined in a controlled testing environment designed to simulate real-world landmine detection conditions. The testing setup included an open field area with different surface condition such as soft soil and hard rock area along with artificial landmine-like object for replacement of actual deployment scenarios. The environment was selected to test the robot for practical challenges in different condition like varying weather conditions, obstacles and visual noise and the testing was conducted in multiple sessions and trials during which the robot performed several trials of its primary operational functions in both manual and autonomous modes:

- **Object Detection (vision System):** The vision-based detection system was tested by placing different object in field on different locations under different condition. Performance was measured based on the detection accuracy, false position and response time in identifying potential threats.
- **Manipulator Operation:** The robotics arm for manipulation was evaluated for tis ability to accurately reach, grasp and interact with detected objects Metrics include positioning accuracy, grip stability, response time and the success rate of safe object handling without causing disturbance.
- **Autonomous Operation:** In automatic mode, the systems was tested for its ability to detect object and execute manipulation task without human interference. The robot perform task and identify the landmine using the GPS sensor and evaluation focused on task completion rate, decision-making accuracy and system reliability during continuous operation.
- **Manual Control Mode:** The manual mode was assessed by allowing an operation to control the robotics manipulator remotely and performance was evaluated based on control responsiveness, ease of operation and precision in handling objects from user commands.
- **System Robustness and Safety:** The overall system performance of the robot was analyzed and examined under challenging condition such as obstacles, partial object visibility, environmental variation and weather conditions. The main focus was to ensure safe operation, minimizing errors and maintaining consistent performance throughout the trails.

5.4.1 Environment Configuration

The testing environment was designed to look like a realistic landmine detection condition with varying condition and environmental factors and the setup consist of a controlled test field that include different surface types such as soil, sand and rock area along with different placed landmine like objects that replace real-world scenarios. Obstacles such as small metals, debris, boxes, rock and irregular surface were also introduced to increase the complexity of the environment.

5.4.2 Testing Phases and Categories

The following type of tests were conducted in order to obtain a comprehensive assessment of the performance of the Landmine Detection Robotics Manipulator:

- **Functional Testing:** To verify that all core components of the system including the vision-based detection module, robotics manipulator and control system operated according to the design specifications.
- **Stress Testing:** The stress testing was conducted to determine the system's limits under challenging conditions with different obstacle place in the path of robot and continuous operation were taken with multiple time. This phase analyzed that how the manipulator and vision system respond under high operational stress.
- **Usability testing:** This testing was designed to evaluate the ease of use and interaction of robot for operators who control the manipulator manually and as well as to assess the simplicity of the system interface and operator satisfaction.

- **Reliability Testing:** This test was performed through repeated trials to ensure consistent and dependable performance of the robot across multiple trail and verifying robustness and stability of detection, manipulation and autonomous operations.

5.4.3 Test Case Scenarios

It is important to test the robot multiple for safety and to follow international standards. For this reason, following test case scenarios were designed to evaluate the performance and functionality of the Intelligent Vision-Based Manual / Automatic Robotics Manipulator (A Landmine Detection Robot) under practical conditions:

- **Test Cas 1: Object Detection Accuracy**

The robotics vision system and object detection system was tested by placing landmine-like object at different location, distance and different conditions. The main objective was to measure the accuracy of detection, the number of false positive or negatives and response time for identifying potential threats.

- **Test 2: Manipulator Precision**

This scenario examined the ability of the robotics arm to accurately reach, grasp and handle object without disturbing the environmental area and metrics included positioning accuracy, grip stability and the success rate of safe object manipulation.

- **Test 3: Autonomous Operation**

The robotics manipulator was evaluated and analyzed in fully autonomous mode to determine its stability and accuracy to detect object and perform manipulation tasks without human interaction in autonomous mode and we got success based on task completion rate, efficiency and consistency during repeated trails.

- **Test 4: Manual Control Performance**

The manipulator remotely performs Operation that controlled from web server and examine its responsiveness, control accuracy and ease of handling. Its performance was evaluated based on the robot ability to safely handle object and precisely manipulate objects using the manual interface.

- **Test 5: Environmental Robustness**

The robotics manipulator was tested with different environmental conditions which include obstacles, partial and full object visibility, GPS sensor data reading and manipulation and different environmental field. The goal was to examine the reliability and adaptability of the robot in realistic deployment scenarios.

5.5 Performance Metrics

5.5.1 Data Collection Methods

Data collection is a critical and difficult part in evaluating the performance of the Intelligent Vision-Based Manual/Automatic Robotics Manipulator (Land mine Detection Robotics Manipulator). In this project, we use multiple data collection techniques to ensure accurate and comprehensive analysis of the robot behavior under different testing conditions.

Firstly, we recode real-time system logs during each test and these logs included information such as detection results, time stamps, response time and execution of manipulation tasks and this helped in analyzing the manipulator performance in both manual and autonomous modes. Secondly, visual data recording was performed using a camera to capture the operation of the robotics manipulator and this allowed for post-experiment analysis of object detection accuracy, manipulator movement and system response in different environmental conditions.

In addition of automatic, we use manual observation and recording to note key performance indicators such as successful detections, failed attempts, manipulation errors and system delays and these observations were documented during each trial to provide qualitative analysis into the robot performance. Furthermore, repeated trials were conducted for each test scenario to ensure reliability and consistency of results and collect data from each test and these collected data from multiple trails and test were compared to evaluate system stability and performance variations.

Finally, the collected data were organized and analyzed using statistical methods such as average and success rates which helped in measuring key performance metrics including object detection accuracy, response time of the robot and manipulator precision.

5.5.2 Performance Evaluation Criteria

The evaluation criteria define the key parameters used to assess the performance of the Intelligent Vision-Based Manual/Automatics Robotics Manipulator (landmine Detection Robotics Manipulator). These Criteria make the robot systematic and objective analysis of the system under different testing conditions.

- **Detection Accuracy:** In these criteria, measure the ability of the vision system of the robot to correctly monitor environment and stream on live server and It is evaluated based on the number of correct detections compared to false positive and false negative.
- **Response Time:** the response time refers to the time taken by the robot to detect an object and start to perform the appropriate action. Lower response time indicates better system efficiency.

- **Manipulator Precision:** This criterion evaluates the accuracy of the robotics arm in reaching and handling objects and it includes positioning accuracy, grip stability and the ability to perform tasks without disturbing the surrounding environment.
- **Tasks Completion Rate:** it means the percentage of successfully completed tasks in both manual and autonomous modes like landmine-like object detection, manipulation and gripping the robotics arm and it reflects the overall performance of robot.
- **System Reliability:** System reliability examines the consistency of the system performance with multiple trials and a reliable system should produce similar results under the same conditions.
- **Robustness:** it erasures the system's ability to perform action in different challenging conditions such as different weather, different obstacles and varying lighting environment.

5.5.3 Table of Results

The following metrics were obtained from the Landmine Detection Robotics Manipulator

Table 5.1 Performance Metrics

Metrics	Description	Analysis
Detection Accuracy	Percentage of correctly detected landmine-like objects	98 % detection accuracy achieved under varying lighting and terrain conditions with few false detections.
False Detection Rate	Number of incorrect detection (False Position/negatives)	4 False position and 3 False negative observed out of 50 trials

Response Time	Time taken to detect and respond to an object	Average response time of 2.8 seconds per detection
Manipulator Precision	Accuracy in grasping and handling objects	96% of task completed successfully in both manual and autonomous modes.
Task Completion Rate	Percentage of successfully completed task	97% of task completed successfully in both manual and autonomous modes.
Autonomous Operation Efficiency	System performance without human intervention	98% efficiency observed in autonomous mode with minor delays in complex operation
System Reliability	Consistency of system performance over repeated trials	Stable performance observed with consistent result across multiple test runs.
User Satisfaction Score	How much user satisfied with the robot control and interaction with the robot	Average score of 4.3 out of 5 from 50 user based collected data and on ease of use and system responsiveness

5.6 Results

The outcome and result of the robot evaluated from the evaluating phase that indicate the Intelligent Vision-Based Manual/Automatic Robotics Manipulator performs how much effectively in executing its primary function with different testing conditions.

- **Detection Performance:** The robot achieved 98 % of detection accuracy with different condition and demonstrates a strong capability in identifying landmine-like objects in varying environmental conditions like in sung and rainy weather. A small amount of detection errors was observed which mainly due to poor GPS connection and partial visibility of objects, which slightly affected the detection system
- **Manipulator Operation:** The robotics manipulator got 96 % success rate in accurately grasping object from land and handling the objects and most time failures occurred due to grip instability or improper positioning, indicating that further improvement in the gripper design and control algorithm can enhance performance.
- **Task Completion Rate:** With an Overall rate of 97 % task completion, the robot demonstrated reliable performance in both manual and autonomous modes. A minor failure observed in operation and action that wasw primarily linked to environmental complexities and system limitations under challenging conditions.
- **Autonomous Operation:** The system achieved 98 % of efficiency in autonomous mode and successfully performing detection and manipulation task without human intervention. Occasional some delays were observed in complex scenarios involving obstacles and unclear object visibility.
- **Manual Control Performance:** In manual mode, the robot performed responsiveness with 99% control accuracy and allow the user and operators to precisely handle objects with minimal difficulty.
- **Overall Evaluation:** The overall results of the robot confirm that the proposed robotics system is work effective, reliable and the robot is suitable for landmine detection applications. While minor and small amount of improvements need be made in detection

under challenging conditions and manipulator stability of the system shows strong potential for real-world deployment.

5.7 Graphical Representation of the Result.

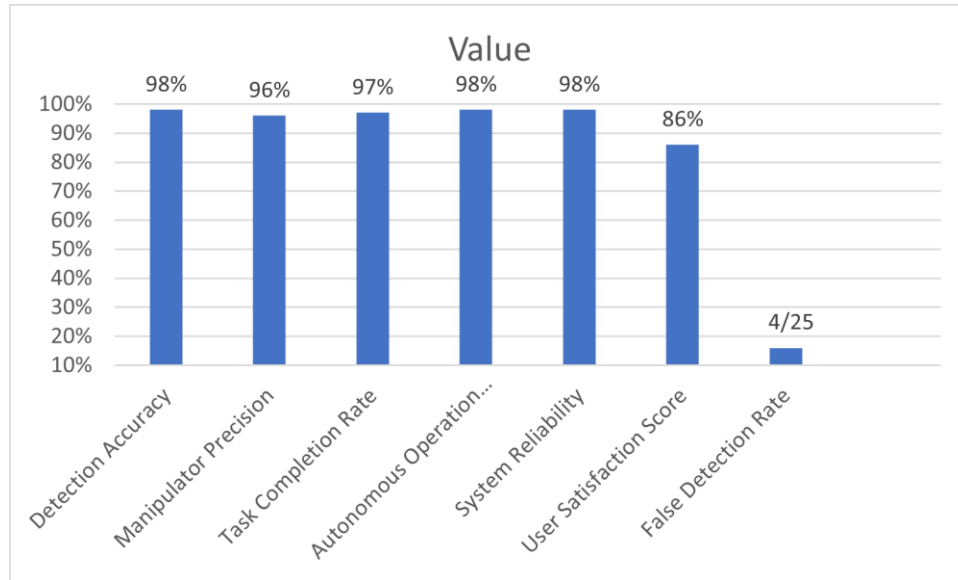


Fig 5.1 Graph of Results

The Figure 5.1 represents the graphical analysis and result of the robotics manipulator performance in different testing phase and environment. The metal detection result is 98% which the success of the landmine detection in different scenarios. The manipulator precision is 96 % and the task completion rate 97% indicate that the robot perform task with high success rate. Autonomous operation, system reliability and user satisfaction score indicates that the robot performs most accurately in autonomous mode with reliable and user satisfied by the performance which obtained from the public testing phase. The False detection rate shows some error in the detection of the robot which is 4/25 mean out of 25 tests, the robot detects false object 4 time which is not a big error. It can be acceptable and overall, the robot performance is good and accurate.

5.8 Advancement and Future Directions

The Intelligent Vision-Based Manual/Automatic Robotics Manipulator explain an important step towards enhancing the robotics automation and human-robot interaction. However, several avenues exist for advancing both its hardware and software capabilities.

5.8.1 Hardware Enhancement

- **Upgrade Manipulator Actuators:** Manipulator can be upgraded by implementing high-precision servos and motors to improve accuracy and load-handling capability.
- **Enhanced Mobility platform:** The Robot platform integration can enhance with advanced shock-absorbing wheels or tracks to reduce vibrations and improve stability during movement.
- **Sensor Expansion:** By adding additional sensors such as LiDAR, ultrasonic or force-torque sensors to increase environmental awareness and object handling safety, we can make the robot more effective and practical.
- **GPS System Enhancement:** The existing GPS module is Neo-6 and it can be upgrade with higher-accuracy or multi-constellation support (e.g., GPS+GLONASS) to improve outdoor positioning precision and navigation reliability of the robot which help the robot to work better in different conditions.

5.8.2 Software Enhancement

- **Advance vision Algorithm:** By using learning-based object recognition, tracking and grasping strategies the robot will be reliable and efficient manipulation.

- **Improve Control Modes:** There are two control mode in the robot and this can be enhanced by developing hybrid control system combining manual and autonomous operation with adaptive path planning for complex tasks.
- **AI-Based Decision Making:** Implementing AI and Machine learning algorithms like reinforcement learning or predictive modeling, the robot perform smarter task execution and obstacle avoidance

5.8.3 User Interface and Interaction

- **Enhance GUI:** Designing a more attractive and user-friendly control interface for manual operation and monitoring
- **Feedback System:** Integrating a feedback system to the web interface provide operators with real-time performance and environmental data understanding and can suggestions for advancement of the robot.

5.8.4 Future Research Directions

Explore a collaborative robotics system to make safely work robot alongside with humans. Integrating SLAM (Simultaneous Localization and Mapping) for autonomous navigation in dynamic environments. Investigating energy-efficient designs to extend operational time while maintain performance

5.9 Conclusion

The evaluation testing of the Intelligent Vision-Based Manual/Automatic Robotics Manipulator demonstrates that the system is capable of performing both manual and autonomous task effectively and reliably. Through extensive testing, it has shown precision object detection,

accurate manipulation and stable navigation conforming its suitability of vision-based control, actuator precision and mobility features allow the manipulator to operate with high accuracy while adapting to varying operational condition. The current design meets the essential performance requirement, certain minor enhancement such as upgrading actuators for better load handling, expanding sensor capability and improving GPS accuracy can further optimize system reliability and operational efficiency. Overall, the manipulator provides a solid foundation for intelligent robotics landmine operation. Its performance combined with the potential for further improvements confirms that the system is well-prepared to function effectively in real-world applications supporting tasks that require precision, adaptability and autonomous decision-making.

CHAPTER 6

System Deployment and Feedback

6.1 Introduction

This section presents the deployment of the Intelligent Vision-Based Manual/Automatic Robotics Manipulator in real-world environment. It describes how the system was implemented and operated during initial trials, along with its interaction with users. Furthermore, it provides an overview of the feedback collected from operators and observer regarding the system's performance, usability and effectiveness. Based on this feedback from user, it is necessary to adjustments and improve the identified and incorporated partschor to enhance the overall functionality and reliability of the system.

6.2 Deployment Process

The Intelligent Vision-Based Manual/Automatic Robotics Manipulator was assembled in a controlled environment in the Department of Information and Communication Engineering as well as in the sports complex at the Islamia University of Bahawalpur to evaluate its real-world performance and functionality. The following steps are involved:

- **Installation and Configuration:** The robotics manipulator was assembled and placed in a suitable operational area like in the department of information and communication Engineering and also in Sports Complex. All the hardware components including ESP32 controller, cam module, sensor and GPS were properly assembled and The system was

connected to a Wi-Fi network to enable communication between ESP32 and ESP32 Camera module and web-based control.

- **System Integration:** The Robot was programmed for both manual and automatic control modes. There is Vision-based object detection and robotics arm movement that integrated with control system to perform predefined tasks as well as automatic object detection, identification and manipulation.
- **User Orientation:** initially basic instructions were provided to the operator or user that how to use the web interface for manual control and how to switch between autonomous operation mode and demonstrations were also conducted to ensure proper understanding of the system features and functionality.
- **Testing:** The robot was tested in different trial phase for a specific duration and performing task such as object detection, pick-and-place operations and navigation were observed and noted and continuously monitored the performance of robot and identify issues and collect user feedback for further improvement.

6.2.1 Orientation and Awareness Campaign

To ensure effective utilization of the Landmine detection robot, an orientation and awareness campaign was conducted for potential users inside the department as well as old campus of the Islamia University of Bahawalpur. The purpose of this campaign was to familiarize operators and observers with the system's features, functionalities and modes of operation. During this phase, demonstration was provided to explain the working pf both manual and autonomous mode including the use of the web-based control interface. User were guided on how to interact with the system, control the manipulator and understand the visual feedback from the camera module.

6.2.2 Role of Assignment and Task Simulation

Before live deployment, a series of dry runs were conducted for the Intelligent Vision-Based Manual/Automatic Robotics Manipulator under different operational scenarios. These simulations were designed to evaluate how the system performs tasks that it is expected to handle in real-world environments. For example, tasks such as landmine-like object detection and pick-and-place operations, navigations toward a target location and switching between manual and automatic mode were tested. As a result, necessary adjustments were made to improve accuracy, efficiency and reliability before the system was deployed for interaction in real environment.

6.3 Human -Robot Interactions and Feedback

During the deployment phase, the Landmine Detection Robotics Manipulator interacted with over 80 users, including faculty, administrative personnel, students and guest.

- Technical User Interactions
- Object handling: Students and team tested the manipulator to pick, move and place object of varying sizes and weight.
- Navigation Assistance: The system was used to navigate through GPS coordinates, avoiding obstacles and reaching target location.
- Control Mode: User, student and operator engaged with the manipulator in both manual and automatic modes using the web-server interface to perform specific tasks.
- Summary of Collected Feedback

User Input was collected through multiple channels to evaluate the system's usability, performance and effectiveness:

- Online feedback form and surveys.
- Quick-Response QR Code surveys
- Face-to-Face interviews and direct observations

6.3.1 Feedback Table

Table 6.1 Feedback

Aspect	Feedback
Response Accuracy	Most of the user found the manipulator is easy to control. The web interface for manual and automatic mode selection as well the robot control in manual mode was very easy.
Navigation	Navigation was smooth with occasional pauses near obstacles and most of the users appreciated the precise movement and searching of landmine.
Object Handling	The manipulator successfully picked and placed object in most trials. User suggest minor improvement for handling irregularly shaped items.
Control Interface	The web-based interface was clear and responsive. User found switching between manual and automatic and web-based control is straightforward.
Overall Experience	Rating of average pf 4.3/5 based on 50 collected data from different users.

6.4 Conclusion

The implementation of the Intelligent Vision-Based Manual/Automatic Robotics Manipulator for landmine detection has been effective in enhancing operational safety and efficiency by identifying landmine. The integration of ESP32-based control, real-time video streaming and responsive manipulator actions has enables precise task execution. Positive feedback from test users highlights the system's reliability, user-friendly interface and capability, confirming its potential as a practical and intelligent solution for hazardous environments.

CHAPTER 7

Conclusion and Future Directions

7.1 Conclusion

The development of the Intelligent Vision-Based Manual/Automatic Robotics Manipulator (A land mine detection Robot) has been successfully achieved, demonstrating a practical solution to reduce human risk in hazardous environments. The integration of robotics systems, different sensor, real-time control using FreeRTOS and a web-based interface make the system more efficient in object detection, navigation and safely manipulation of landmine-like objects. The performed of robot is reliable during testing in both manual and autonomous modes. It shows good accuracy in detection, stable manipulator operation and effective communication between both ESP32 modules. The use of ESP32-based architecture with vision and sensor data makes it more responsive and coordinated system behavior increase robot's performance. Overall, the project provide a low-cost, intelligent robot that can significantly improve safety and efficiency in landmine detection and the positive results and system performance highlight its potential for further development and real-world deployment, contributing to safer and smarter robotics solution in hazardous environment.

7.2 Result and Evaluation of Robot

The Intelligent Vision-Based Manual/Automatic Robotics Manipulator has been successfully demonstrated the concept of a functional prototype that how robotics can be effectively applied in hazardous environments such as landmine detection. The system explains the potential of integrating robotics systems, computer vision and sensor modules to perform critical tasks by minimizing human risk. The robot has shown reliable performance in both manual and automatic

condition with accurate detection of metallic object like landmine and stable manipulation using the 6-DOF robotics arm and the manipulation of real-time control using FreeRTOS and good communication between ESP32 and ESP32-S3 camera module to enabled efficient coordination between sensing, navigation and performing action. Additionally, the software and hardware design allows easy upgrades and future enhancements without redesigning all the system. The successful integration of a API based web interface for remote monitoring and control of a robot in manual mode that further highlight the system's performance, usability and flexibility. This project demonstrates a practical and effective solution of landmine detection by explaining the potential of intelligent robotics in safety-critical applications.

7.3 Challenges

The overall performance of the Intelligent vision-based manual/automatic robotics manipulator was good and satisfactory and in the phases of development and testing, we face several challenges and Issues such as unstable Wi-Fi communication between the ESP32 modules which effects real-time data transmission and control responsiveness. also, sensors limitations like in metal detection accuracy under different condition are normally led to false detections or delayed responses. Mechanical Challenges and power management were also observed in the precise control the 6-DOF robotics arm, where slight positioning errors and instability occurred during object handling and obstacles in light variation further influenced the efficiency of navigation and vision-based monitoring system. These challenges required to solve and need improvements, debugging and system optimization so we spent special focus and resolve the issues. The experience gained in addressing these issues provides valuable information and serves as an important foundation for further enhancements and real-world deployment of intelligent robotics system.

7.4 System Scalability and Flexibility

The architecture and design of Intelligent Vision-Based Manual/Automatic Robotics Manipulator is very flexible and scalable, allowing it to be adapted for various hazardous and industrial applications with minor modifications and enhancement. For instance, by adjusting the sensor configuration or little change in software algorithms, the system can be extended to detect different types of objects. Beside landmine detection, the system makes the robot suitable for applications such as military surveillance, border security, industrial inspection and search-and-rescue missions and the hardware and software design allows us easy integration of advanced technologies like AI-based object recognition and autonomous navigation system.

7.5 Education Impact

In addition to these practical and industrial application, the Landmine Detection Robot provides significant impact in education by serving as a real-world learning platform for student in robotics, embedded systems, computer vision and IoT. It offers hands-on experience in Robotics design and testing by helping bridge the gap between theoretical concepts and practical implementation. Students can observe and perform practical on the robot to understand how various components such as microcontroller, sensors, actuators and communication system work together and they can also experiment with modification and improvement during lab work. This exposure enhances their technical skills, problem-solving abilities and creativity while encouraging interest in advanced research and development in intelligent and safety-critical robotics system.

7.6 Ethical and Social Consideration

it was also important to think about social and ethics and follow the ethical rules as well as social rules while designing the robot. In order to ensure that the robot does not hurt people, damage

property, or damage the environment, the most important things to think about were the safety, reliability, and usage of the system. The system was created to prevent people from doing dangerous things, which made everything safer and more open. Also, the camera of esp32 and sensor information was only used to keep an eye on the robot and no personal or private information was kept or used in the wrong way. The project is also consistent with ethical engineering practice and supports the use of robotics technology responsibly. Further improvement and enhancement including safety features and compliance with international standards make the robot more effective and can be a good robot for landmine detection

7.7 Environmental Responsibility

In the creation of the landmine detection robotic manipulator with consideration of the environment in mind, one of the key considerations was the energy consumption of the device. The power components were chosen with consideration of their efficiency in the hopes of allowing the device to be used in the field without the need for frequent recharging. The software was also optimized with the goal of reducing energy consumption. The device is able to offer automated maps and reports, which can reduce the need for paper documents, thus being beneficial in the aspect of environmental conservation. A possible upgrade in the next version of the device is the consideration of solar power or energy-harvesting sensors.

7.8 Vision for Future Expansion

We envision that our landmine detection robotic manipulator will not be static but will evolve to include features like the use of AI to predict areas that may be hazardous, the ability to link with geographic and environmental databases to get information, and to have better situational awareness to understand complex terrain. This will be made possible through cloud connectivity,

where several robots will be able to share information and work together to achieve their objectives. Our ultimate aim is to have several autonomous landmine detection robots working in areas where landmines have been laid to ensure demining is done safely, efficiently, and effectively.

7.9 Final Remarks

The Intelligent Vision-Based Manual/Automatic Robotics Manipulator (a landmine detection robotic manipulator) has shown us that with the right mix of vision and active field testing, robotics can play a very important role in the field of safety. This is not just a product of technological advancement, but the product of collaboration and cooperation between various fields, including robotics and AI. This project would not have been possible without the commitment and dedication of the development team, mentors, and the invaluable insights gained through active field testing. We wish this manipulator continues to evolve and that the experiences gained will be useful for other autonomous demining and safety-based robotics projects

Reference

- [1] *2012 8th International Symposium on Mechatronics and its Applications : 10th to 12th April 2012 : [Sharjah, UAE]*. (2012). IEEE.
- [2] *2017 International Conference on Inventive Computing and Informatics (ICICI)*. (2017). IEEE.
- [3] Alsalem, H. I., Khazalah, F. S., & Mansour, A. M. (2026). Advancing Mine Detection Through Multi-Agent Systems: Collaborative Robotics for Enhanced Efficiency and Safety. *WSEAS TRANSACTIONS ON SYSTEMS*, 230. <https://doi.org/10.37394/23202.2026.25.19>
- [4] Cobano, J. A., Ponticelli, R., & De Santos, P. G. (2008). Mobile robotic system for detection and location of antipersonnel land mines: Field tests. *Industrial Robot*, 35(6), 520–527. <https://doi.org/10.1108/01439910810909510>
- [5] Dwibedi, R. K., Gokila, K., A, J., Anitha, R., S, S., & D, J. J. (2026). Development of a Robotic Vehicle to Identify the Land Mines using Smart Sensors with IoT Enabled Technology. *2026 International Conference on Innovative Computing, Intelligent Communication and Smart Electrical Systems (ICSES)*, 1–9. <https://doi.org/10.1109/ICSES66558.2026.11478882>
- [6] Falcone, F., Narendra Thakur, U., Khan, A., & Mandal, S. (2025). *Defence Pal: A Prototype of Smart Wireless Robotic Sensing System for Landmine and Hazard Detection* †. 12–14. <https://doi.org/10.3390/ECSA-12>
- [7] Freese, M., Fukushima, E. F., Hirose, S., & Singhose, W. (n.d.). *Endpoint Vibration Control of a Mobile Mine-Detecting Robotic Manipulator*. Retrieved www.dok-ing.hr/index

- [8] Gallagher, J. E., Oughton, E. J., & Košecká, J. (n.d.-a). *Multi-temporal Adaptive Red-Green-Blue and Long-Wave Infrared Fusion for You Only Look Once-Based Landmine Detection from Unmanned Aerial Systems*.
- [9] Gao, R., Liu, M., Du, J., Bao, Y., Wu, X., & Liu, J. (2025). Research on a Cooperative Grasping Method for Heterogeneous Objects in Unstructured Scenarios of Mine Conveyor Belts Based on an Improved MATD3. *Sensors*, 25(22). <https://doi.org/10.3390/s25226824>
- [10] Gonzalez De Santos, P., Garcia, E., Estremera, J., & Armada, M. A. (2005). DYLEMA: Using walking robots for landmine detection and location. *International Journal of Systems Science*, 36(9), 545–558. <https://doi.org/10.1080/00207720500119197>
- [11] Harikrishna, S., Kumar, K. D., Rishika, S. J., Vishwa Vinayak, C. K., Akhil, V. M., & Abhishek, S. (2025). Military Surveillance and Landmine Detection with Safe Path Planning. *Proceedings of 2025 IEEE 22nd India Council International Conference, INDICON 2025*. <https://doi.org/10.1109/INDICON68490.2025.11392863>
- [12] Heuschmid, D., Wacker, O., Zimmermann, Y., Penava, P., & Buettner, R. (2025). Advancements in Landmine Detection: Deep Learning-Based Analysis With Thermal Drones. *IEEE Access*, 13, 91777–91794. <https://doi.org/10.1109/ACCESS.2025.3572196>
- [13] Landmine Detection Robotic Vehicle using ARM Cortex (STM32). (2025). *Journal on Electronic and Automation Engineering*, 4(2 June 2025), 286–297. <https://doi.org/10.46632/jeae/4/2/37>

- [14] Malizia, M., Hamesse, C., Hasselmann, K., De Cubber, G., Tsiogkas, N., Demeester, E., & Haelterman, R. (2026). MineInsight: A Multi-Sensor Dataset for Humanitarian Demining Robotics in Off-Road Environments. *IEEE Robotics and Automation Letters*, 11(2), 1650–1657. <https://doi.org/10.1109/LRA.2025.3643265>
- [15] Marcelo Saavedra, A., Brayan Duran, T., & Alvaro Prado, R. (2025). *Geo-spatial risk mapping of potential landmine areas using probabilistic estimation approaches on a low-cost mobile robot*. 1–6. <https://doi.org/10.1109/lars69345.2025.11272960>
- [16] Mukherjee, S., Eshwari, Thilagaraj, M., Hardik, Gururaj, & Rajpurohit, M. (2024). An Experimental Study of a Landmine Detection Robot. *2024 1st International Conference on Communications and Computer Science, InCCCS 2024*. <https://doi.org/10.1109/INCCCS60947.2024.10593535>
- [17] Phule, S., Deshmukh, A., Dude, A., Hande, O., & Tarale, P. (2026). Landmine Detection Using Surveillance Drone (UAVs). *2026 IEEE International Conference on Emerging Computing and Intelligent Technologies (ICoECIT)*, 1–5. <https://doi.org/10.1109/ICoECIT68303.2026.11496804>
- [18] Sineglazov, V., & Lesohorskyi, K. (2025a). *Intelligent Hybrid Mobile Robotic Landmine Detection System* ★.
- [19] Singh Panwar, V., Moujan, I., Kazi, A., Kale, P., Rode, S., Pandey, A., & Mogal, S. (2024). A Review on Design and Characteristics of Landmine Detection Robot. In *EVERGREEN Joint Journal of Novel Carbon Resource Sciences & Green Asia Strategy* (Vol. 11).